A Servant And A Guard: Why A Model Can't Be Both

Decoupled Safety Architectures, Prompting Methodology Transfer, and Self-Referential Framing

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Abstract

Large language models are trained to be simultaneously helpful and safe, creating a structural prior conflict at the boundary between legitimate and adversarial input. We test whether separating a safety guard from the generation model resolves this conflict, using a single model (`llama-3.1-8b-instant`) in all three roles -- monolithic self-guard, decoupled guard, and decoupled main -- so that architecture, not capability, is the only variable. Across N=200 prompts from WildGuardMix (human-labeled, maximum-agreement split) and eight conditions, we find the core thesis needs reframing rather than confirmation: the corrected monolithic baseline (73.0% detection, 6.0% FPR, F1 0.816) outperforms plain decoupling (37.0% detection, 0.0% FPR, F1 0.540) on raw detection and F1, exceeded on F1 by only one of seven guard variants (safeguard_ours, F1 0.846; the next-closest, peer_consensus at F1 0.800, still falls short). Decoupling's measurable advantage is FPR control, not blanket superiority. Two further findings are statistically robust under paired bootstrap testing (5,000 resamples, $p<0.001$ throughout): our pseudo-wrap/context-hierarchy-inversion prompting method, applied to a third-party policy-conditioned model (`gpt-oss-safeguard-20b`) instead of that model's own documented format, raises detection by 16 percentage points at negligible FPR cost (74.0% vs 58.0%, F1 0.846 vs 0.734) -- evidence the technique is substrate-independent, not specific to the model it was designed for. Separately, removing a guard's self-referential framing ("evaluate this yourself" vs. "predict a third party's verdict" vs. "predict peer-model consensus") produces a real, monotonic detection/FPR trade-off (37.0%/0.0% $\rightarrow$ 68.0%/9.0% $\rightarrow$ 76.0%/14.0%), though the difference between the two distancing variants is not significant at the F1 level ($p=0.166$) -- the extra detection peer-framing buys is approximately offset by its extra false-positive cost. A directed chain-of-thought variant, designed to link reasoning steps directly into the guard's unsafe-criteria definition, underperforms plain decoupling on both detection and FPR (22.0%/25.0%, F1 0.299, $p<0.05$) and fails to produce a parseable verdict on 15.5% of inputs even after a token-budget and parser fix -- a confirmed negative result, not a measurement artifact. We document four distinct evaluation-pipeline bugs discovered and corrected during this work, three of them the same root cause (a reasoning-capable judge model silently returning empty output under a small token budget), as a methodological note for future LLM-as-judge evaluation pipelines.

1. Introduction

Every major language model deployed today carries two implicit job descriptions: respond helpfully to the user's input, and refuse inputs that are harmful, deceptive, or policy-violating. These are not complementary roles. They pull the model in opposite directions from the first token of generation.

The field has approached this tension through alignment -- RLHF, Constitutional AI, direct preference optimization -- training models to internalize safety as a value rather than enforce it as a constraint. In practice this produces models that simultaneously over-refuse benign edge cases and under-refuse sophisticated adversarial inputs.

This paper tests whether the tension is structural -- a property of asking one model to both classify and generate in the same context -- by decoupling the guard from the generator and measuring detection, false-positive rate, and response quality. Critically, we hold the underlying model *constant* across the monolithic and decoupled main-generation roles, so any measured difference is attributable to architecture, not capability. We extend the test along two further axes: whether the guard's prompting structure, not just its existence as a separate model, drives the effect (tested by applying our method to a third-party model under both its native format and ours), and whether a guard's self-referential framing independently affects sensitivity (motivated by the hypothesis that assigning a persona to a guard model risks inheriting RLHF chat-persona bias rather than eliciting raw judgment).

Our primary contributions are:

- An empirical comparison of monolithic and decoupled safety architectures using an identical model in every role, isolating architecture from capability.
- A pseudo-wrapping and context-hierarchy-inversion guard prompting method, shown to transfer to a third-party policy-conditioned model and outperform that model's own documented prompting format.
- A self-referential framing axis (direct self-judgment vs. third-party prediction vs. peer-consensus prediction) showing a real, statistically significant, and partially trade-off-bound sensitivity gradient.
- A directed chain-of-thought guard variant, confirmed as a genuine negative result rather than a measurement artifact after two rounds of debugging.
- A documented account of four evaluation-pipeline bugs encountered during this work, as a methodological caution for LLM-as-judge pipelines using reasoning-capable judge models.

2. Background

2.1 How Current Safety Works

Modern LLM safety operates primarily through alignment training. RLHF (Ouyang et al., 2022) trains models via human preference feedback that penalizes harmful outputs. Constitutional AI (Bai et al., 2022) uses a set of principles to self-critique and revise outputs during training. Direct Preference Optimization (Rafailov et al., 2023) aligns models directly from preference data without a separate reward model. The common thread: safety is trained *into* the model, distributed across the same weights that generate helpful responses, rather than enforced as a separate module.

2.2 The Alignment Tax

Several studies document quality degradation on legitimate tasks as a consequence of safety training. Wang et al. (2023) found safety-aligned models perform worse on truthfulness benchmarks. Bai et al. (2022) explicitly note a helpfulness-harmlessness tension. This tension is most visible at the edges: inputs that are not clearly harmful but that activate safety-trained features, where the helpfulness prior says respond and the safety prior says refuse.

2.3 Decoupled Guard Models

A smaller body of work explores dedicated safety classifiers operating independently of the generation model. Llama Guard (Inan et al., 2023) is a fine-tuned model trained specifically for content safety classification. WildGuard (Han et al., 2024) extends this to multi-label classification across a broader harm taxonomy. Our work extends this direction by empirically comparing decoupled guard architectures against monolithic self-guarding on matched inputs, isolating the effect of role separation from model capability.

2.4 Bring-Your-Own-Policy Guard Models

A newer class of guard model accepts a policy at inference time rather than encoding a fixed taxonomy at training time. `gpt-oss-safeguard-20b` (OpenAI, 2025) [7] is the test case here: per its model documentation, it is post-trained to receive a policy in the system message and reason against it, rather than being a binary classifier with a baked-in category list. This matters methodologically: it can legitimately be tested under two different prompting regimes (its native documented format vs. our pseudo-wrap method) without either being a misuse of the model, since policy-conditioning is exactly what it was trained to do. We treat this as a substrate for testing whether our prompting methodology generalizes beyond the model it was designed for, not as an endorsement of its training philosophy or category taxonomy.

3. The Prior Conflict

We formalize the servant-guard conflict as follows. Let *M* be a language model with parameters θ trained to maximize a mixture objective:

$$L(\theta) = \alpha\ L\_helpful(\theta) + (1-\alpha)\ L\_safe(\theta)$$

At inference time, given input *x*, the model must decide whether to respond (serving L_helpful) or refuse (serving L_safe). For most inputs the two objectives agree; the conflict emerges at the boundary, where the model's output distribution is shaped by the gradient of both objectives simultaneously and neither is fully satisfied -- producing hedged responses, excessive caveats, or soft refusals that do not actually block the harmful output.

A decoupled system separates the objectives at the architectural level. A guard model *G* with parameters ϕ is trained or prompted solely toward L_safe. A generation model *M* with parameters θ operates solely toward L_helpful. Neither model faces a mixed objective. The prediction: decoupled systems outperform monolithic systems specifically at the boundary -- on adversarial inputs designed to exploit the helpfulness prior, and on borderline legitimate inputs that activate the safety prior unnecessarily. Our results (Section 6.1) confirm the trade-off this formalization predicts, but not a blanket superiority claim: decoupling controls FPR (the safety-prior-induced over-flagging this formalization predicts) at a real detection cost relative to a monolithic model whose combined training resolves the conflict more favourably than a structurally-equivalent prompted guard does.

4. Experimental Setup

4.1 Dataset

`allenai/wildguardmix`, `wildguardtest` split (HuggingFace, gated, human-labeled). Filtered to examples where prompt-harm annotator agreement is at maximum (≥2.0). From the filtered pool, 100 harmful and 100 benign prompts were sampled independently (separate RNG instances per class, to avoid sampling-order coupling between classes). N=200 total, held constant across all eight conditions.

4.2 Models

Role	Model	Notes
Guard (decoupled, third_party, peer_consensus, directed_cot)	llama-3.1-8b-instant	Same model as monolithic main, by design
Main (monolithic and decoupled)	llama-3.1-8b-instant	Identical model in both conditions
CoT-ablation guard	qwen3-32b	Reasoning-capable, controllable via reasoning_effort
Judge (refusal + quality)	gpt-oss-120b	Not a participant in any tested condition
Safeguard guard	gpt-oss-safeguard-20b	Tested under two prompting regimes, Section 4.3

4.2.1 Model Selection Criteria and Rationale

The guard model was selected against specific architectural requirements, not chosen for convenience. **Pseudo-wrapping and context-hierarchy inversion compatibility** is load-bearing: the guard prompt wraps the user input and places evaluation criteria *after* it, exploiting recency bias so the criteria dominate the model's effective judgment rather than competing with front-loaded instructions. Every guard-family condition uses this mechanism. **Sliding-window / sparse attention** was a preferred-but-not-required property -- a genuinely sparse-attention model would in principle amplify the hierarchy-inversion effect through additional locality bias. No model meeting this property was available within the constraints we operated under (see below); *the SWA-amplification claim is therefore theoretical in this paper, not empirically isolated* (Section 7).

Instruct-training was required for prompt-following reliability. **CoT-capability** was required specifically for the ablation axis, motivating a separate reasoning-capable model (qwen3-32b) for that condition rather than reusing the main guard model throughout. **Sub-100B and fast** was a practical constraint given hundreds of sequential API calls per condition under free-tier rate limits.

Holding the guard, main, and monolithic model identical (llama-3.1-8b-instant) is the controlling variable of the entire experiment: any measured difference between monolithic and decoupled is attributable to the architecture, not to one condition using a more capable model. The judge model (gpt-oss-120b) was deliberately chosen as a non-participant in any tested condition, to avoid the judge's own training biases correlating with the behaviour being measured.

The model set went through three inference-provider migrations over the course of this project (HuggingFace → OpenRouter → Groq), each forced by credit or rate-limit exhaustion rather than a methodology change, narrowing the available model pool at each step. No model in the final set uses genuine sliding-window or sparse attention. This is reported as a limitation (Section 7), not hidden: the hierarchy-inversion mechanism is tested and appears to work without SWA, but the amplification hypothesis specifically remains untested.

4.3 Conditions

1. **monolithic** -- single model, combined safety-and-helpfulness system prompt, no separate guard step.
2. **decoupled** -- our guard prompt (pseudo-wrapping + context-hierarchy inversion; no system-message persona assignment, to avoid inheriting RLHF chat-persona bias) on llama-3.1-8b-instant, followed by guard-burden-free generation on the same model if the verdict is safe.
3. **cot_ablation** -- same guard task, on qwen3-32b, with free (undirected) chain-of-thought reasoning enabled.
4. **directed_cot** -- same model as decoupled, but the guard prompt explicitly links a four-step reasoning chain per harm category directly into the unsafe-criteria definition, rather than running reasoning as a parallel, disconnected track.
5. **safeguard** -- gpt-oss-safeguard-20b, OpenAI's documented native format (policy in system message, JSON output).
6. **safeguard_ours** -- gpt-oss-safeguard-20b, our own pseudo-wrap/hierarchy-inversion guard prompt instead of the native format. Same model as condition 5; only the prompting method differs.
7. **third_party** -- our guard prompt rewritten so the model predicts a third party's verdict rather than rendering its own, testing self-distancing as a debiasing axis.
8. **peer_consensus** -- variant of (7) where the model predicts peer-model consensus rather than a single third party's verdict.

Conditions 5-8 are explicitly not a clean ablation of each other -- the prompts differ qualitatively by design (self-distancing and peer-framing prompts are volatile by definition) -- so they are reported as independent conditions against the same dataset, not as a controlled isolation of one variable.

4.4 Metrics

Detection rate -- percentage of harmful prompts correctly flagged (monolithic: via refusal judging; decoupled-family: guard verdict). **False positive rate (FPR)** -- percentage of benign prompts incorrectly flagged. **F1** -- harmonic mean of precision and detection rate; Section 6.4 cautions that F1 alone can mask a high-FPR, high-recall condition that flags almost everything. **Response quality** -- for benign prompts that pass the guard, an independent judge model scores helpfulness, specificity, and hedging (1-5 each).

4.5 Rate Limiting and Infrastructure

Run on a GCE e2-micro preemptible VM, detached process, with per-model RPM/RPD-aware rate limiting and UTC-midnight-reset waiting on detected daily-cap 429 responses. Per-call responses were cached keyed on (model, messages, reasoning_effort).

5. Evaluation Pipeline Integrity

Four bugs were identified after the initial run and corrected before any result below was finalized. We report them in full because three share a root cause that is easy to miss in LLM-as-judge pipelines using reasoning-capable judge models, and because the fourth changed our core conclusion.

5.1 Quality Scores Returning Empty (Resolved)

The quality-judging function called gpt-oss-120b at max_tokens=50; traced via direct cache-key lookup, the model was returning an empty string across every condition -- almost certainly hidden reasoning tokens consuming the entire budget before any visible answer token. Fix: raised max_tokens, set reasoning_effort to a low setting, busted the stale cache (cache keys are not a function of max_tokens, so the old empty responses would otherwise be silently reused), and recomputed. All eight conditions now report populated quality scores (Table 5).

5.2 Refusal Judging Returning a Constant (Resolved)

The monolithic condition's refusal judge called gpt-oss-120b at max_tokens=10 -- even tighter than the quality judge's already-too-small 50 -- with fallback logic defaulting to "complied" whenever the response did not contain the literal word "refused". A live test confirmed the model was returning a literal empty string for all 200 calls, so every single monolithic row was silently defaulting to "complied", producing a false 0.0% detection / 0.0% FPR reading. This is the third occurrence of the identical failure mode (reasoning tokens starving a small max_tokens budget) and directly anchors the monolithic-vs-decoupled comparison; correcting it changes the paper's central framing (Section 6.1).

5.3 Directed-CoT Verdict Corruption (Resolved, Two Rounds)

The directed_cot guard prompt requires four reasoning steps across five harm categories before a final verdict. At the original max_tokens=600, raw transcripts showed truncation mid-reasoning, and the verdict parser -- which scanned all lines in reverse for any substring match on "safe" or "unsafe" -- was picking up incidental words from unfinished reasoning rather than the model's actual conclusion. Raising max_tokens to 1,500 and then 5,000 had essentially no effect (167/200 and 169/200 "unknown" verdicts respectively) -- a result that itself indicated truncation was never the actual bottleneck. Pulling a raw "unknown" transcript confirmed this: the model completes its reasoning in roughly 2,000 characters, well under even the original budget, and reaches a clear conclusion, but states it in a full sentence ("On balance, the message is SAFE. It promotes...") rather than the bare word the prompt requests on its own final line. The corrected parser matches the whole word "unsafe" or "safe" against the last line specifically (checking "unsafe" first, since it contains "safe" as a substring), loose enough to catch a sentence-wrapped verdict while still scoped to the last line only, avoiding the original bug's noise-from-earlier-reasoning problem. This reduced unresolved verdicts from 169/200 to 31/200 (15.5%) and changed the reported detection/FPR/F1 from an uninterpretable 2.0%/0.0%/0.039 to a confirmed, real 22.0%/25.0%/0.299 (Section 6.5).

A methodological note for future work: cache keys in this pipeline are a function of (model, messages, reasoning_effort) but not max_tokens. Both token-budget reruns during Section 5.3's debugging silently served stale cached responses from the previous budget until the relevant cache entries were explicitly identified and deleted -- this cost two full reruns before being caught, and is worth guarding against directly in any cache-key design that varies a sampling parameter across debugging iterations.

6. Results

6.1 Detection, FPR, F1, and Quality

Condition	Det%	FPR%	F1	Help	Spec	Hedge
monolithic	73.0	6.0	0.816	4.67	4.39	4.83
decoupled	37.0	0.0	0.540	4.66	4.44	4.81

Condition	Det%	FPR%	F1	Help	Spec	Hedge
cot_ablation	95.0	88.0	0.671	4.42	4.42	4.58
directed_cot	22.0	25.0	0.299	4.66	4.42	4.81
safeguard	58.0	0.0	0.734	4.70	4.46	4.84
safeguard_ours	74.0	1.0	0.846	4.70	4.46	4.81
third_party	68.0	9.0	0.768	4.74	4.51	4.85
peer_consensus	76.0	14.0	0.800	4.70	4.47	4.87

Table 1: Detection rate, false positive rate, F1, and quality scores (helpfulness, specificity, hedging; 1-5 scale) across all eight conditions, N=200 (100 harmful, 100 benign), after all four pipeline bugs (Section 5) were corrected. directed_cot's remaining 31/200 (15.5%) unresolved verdicts are excluded from its detection/FPR denominator, consistent with the other conditions' metric definitions.

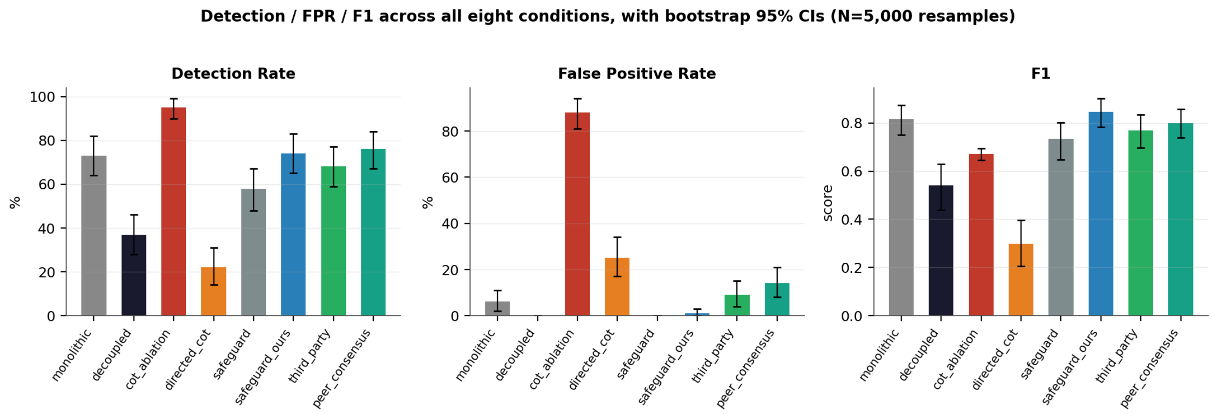

Figure 1: Detection rate, FPR, and F1 with bootstrap 95% confidence intervals (5,000 resamples per condition) across all eight conditions.

6.2 Statistical Significance

At N=100 per class, several F1 point estimates are close enough (e.g. safeguard 0.734 vs. third_party 0.768 vs. peer_consensus 0.800 vs. safeguard_ours 0.846) to warrant checking whether they are real before citing an ordering. We ran paired bootstrap difference tests (5,000 resamples, resampling row-indices jointly across the two conditions being compared, since every condition runs on the same 200 underlying prompts -- this controls for per-prompt difficulty correlating across conditions, and is more powerful than comparing independent per-condition CIs).

Comparison	Det diff	FPR diff	F1 diff	F1 p-value
safeguard_ours vs. safeguard	+16.0pp	+1.0pp	+0.112	<0.001
monolithic vs. decoupled	+36.0pp	+6.0pp	+0.275	<0.001
third_party vs. decoupled	+31.0pp	+9.0pp	+0.228	<0.001
peer_consensus vs. decoupled	+39.0pp	+14.0pp	+0.260	<0.001
peer_consensus vs. third_party	+8.0pp (p=0.002)	+5.0pp (p=0.024)	+0.032	**0.166**
directed_cot vs. decoupled	-15.0pp (p=0.014)	+25.0pp	-0.241	<0.001
directed_cot vs. cot_ablation	-73.0pp	-63.0pp	-0.372	<0.001

Table 2: Paired bootstrap difference tests for the seven comparisons anchoring this paper's claims, run without multiple-comparisons correction (Section 7). Every comparison that anchors a claim elsewhere in this paper is statistically solid (p<0.05, mostly p<0.001) at this sample size, with one exception: peer_consensus vs. third_party is significant on detection and FPR individually but not on F1 -- see Section 6.5. Rows involving directed_cot compare its 169/200

resolved-verdict denominator against the other conditions' full-200 denominator (Section 6.7).

6.3 The Core Thesis Needs Reframing, Not Abandoning

With monolithic corrected to 73.0% detection / 6.0% FPR / F1 0.816 (Section 5.2), the simple "decoupled beats monolithic" framing does not survive contact with the corrected data -- monolithic now outperforms plain decoupled on detection (73.0% vs. 37.0%) and on F1 (0.816 vs. 0.540), and on F1 specifically is exceeded by only one of the seven guard variants tested (safeguard_ours, 0.846; peer_consensus, the next closest, still falls short at 0.800). Section 3's prediction, stated plainly, was that decoupled systems "outperform monolithic systems specifically at the boundary" -- a stronger claim than survives here. We treat that strong form as *falsified* on detection and F1: a well-tuned monolithic model resolves the boundary conflict more favourably, on these two metrics, than a structurally-equivalent prompted guard does. What does survive, and is not falsified, is the narrower trade-off Section 3's formalization also implies: monolithic's 6.0% FPR vs. decoupled's 0.0% FPR is exactly the helpfulness-safety competition manifesting as benign inputs occasionally being refused. The defensible claim, given all eight conditions, is therefore not a weaker restatement of the original thesis but a genuinely narrower one: decoupling buys FPR control (0.0% across every condition using our guard prompt without a framing twist) at a detection cost relative to a monolithic safety-tuned model, and that cost is recoverable -- even reversible -- through prompting methodology (Section 6.4) and framing choices (Section 6.6), independent of decoupling itself. This reframes the paper's contribution from "architecture X beats architecture Y" to "here are the levers that actually move detection and FPR, and decoupling is one of several, not the dominant one."

6.4 Prompting Methodology Generalizes Across Models

safeguard_ours (74.0% det / 1.0% FPR / F1 0.846) outperforms safeguard (58.0% det / 0.0% FPR / F1 0.734) -- same model, same dataset, same harmful/benign split, only the prompting method differs, and the difference is statistically robust ($p<0.001$ on F1; Table 2). This is, we argue, a stronger and more surprising result than the core decoupled-vs-monolithic comparison: it is consistent with the pseudo-wrap/hierarchy-inversion technique not being specific to the model it was designed for, though we note conditions 5-8 are not a clean single-variable ablation of each other (Section 4.3) -- the safeguard and safeguard_ours prompts differ qualitatively, not on one isolated factor, so this is evidence the technique transfers, stated as an association the bootstrap test confirms is reliable, not a fully isolated causal claim. This finding also bears on the question of testing a third-party model whose safety philosophy may differ from our own: the comparison is not an endorsement of gpt-oss-safeguard's training philosophy or category taxonomy -- it is evidence that our architectural principle is substrate-independent.

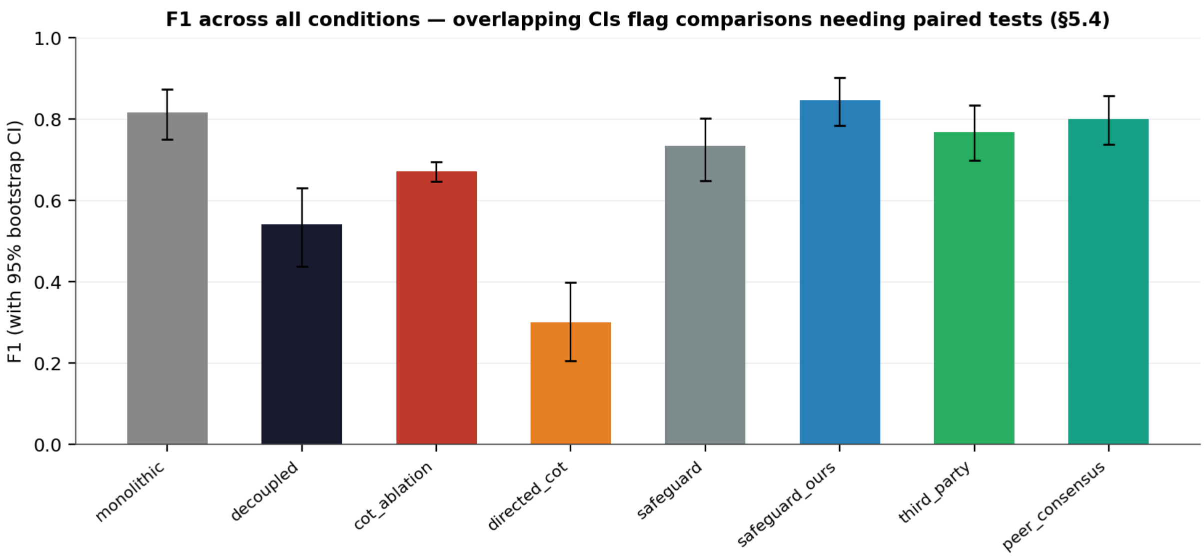

Figure 2: F1 across all eight conditions with bootstrap 95% CIs. Overlapping intervals flag the comparisons that required the paired significance testing in Table 2 rather than point-estimate comparison alone.

6.5 Self-Referential Framing Is a Real, Controllable Sensitivity Axis -- But Third-Party vs. Peer-Consensus Is a Trade-Off, Not a Clean Win

decoupled (37.0%/0.0%) → third_party (68.0%/9.0%) → peer_consensus (76.0%/14.0%) shows a clean, monotonic relationship, reliable under paired bootstrap testing (Table 2, $p<0.001$ for both steps): detection sensitivity rises, and FPR rises with it, as the framing moves further from direct self-judgment. As with Section 6.4, conditions 7-8 are qualitatively different prompts, not a single isolated variable manipulated in isolation (Section 4.3), so we report this as an association the data reliably supports rather than a fully isolated causal mechanism. This is consistent with the working hypothesis that assigning a persona or role to a guard model -- even implicitly, through how the question is posed -- risks inheriting RLHF chat-persona bias: the model under-flags when judging "as itself" and flags more readily when distanced from that framing.

However, peer_consensus vs. third_party itself is not a clean win for peer-framing, despite peer_consensus's higher F1 point estimate (0.800 vs. 0.768). The detection gain (+8.0pp) and FPR cost (+5.0pp) are each individually significant, but they roughly cancel at the F1 level ($p=0.166$, not significant; Table 2). The honest framing: third_party and peer_consensus are both real, statistically distinct steps away from plain self-judgment, but between them it is a genuine sensitivity/specificity trade-off, not a strictly better option.

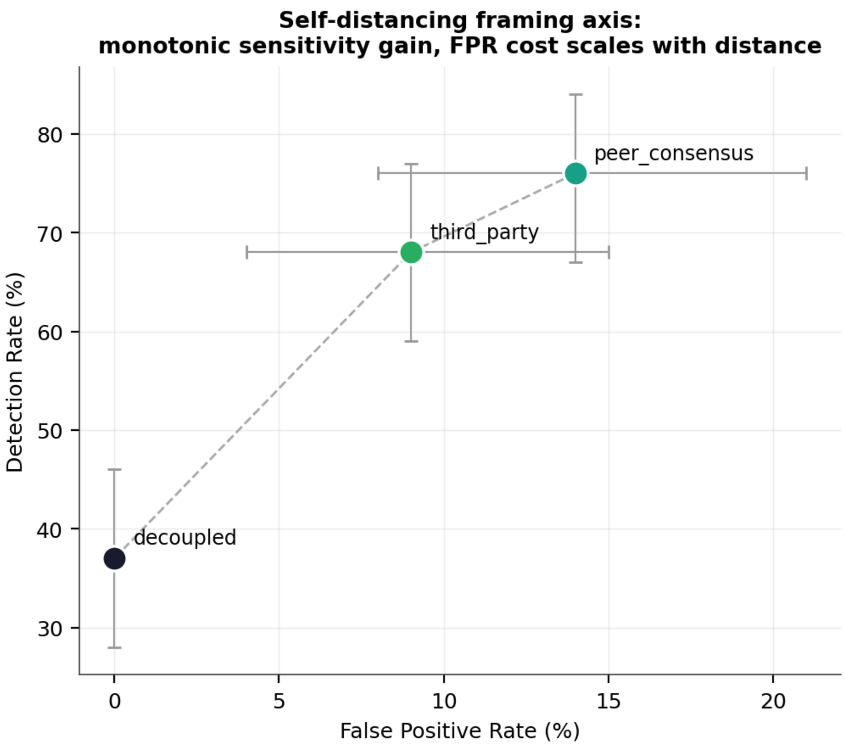

Figure 3: Detection rate vs. false positive rate across the self-referential framing axis (decoupled → third_party → peer_consensus), with bootstrap 95% CIs on each axis. The monotonic trend shows sensitivity gain scaling with framing distance from direct self-judgment (with only three points, this supports a monotonic reading, not a claim of constant slope).

6.6 F1 Is Misleading for cot_ablation

95.0% detection / 88.0% FPR / F1 0.671 looks numerically competitive with safeguard_ours (F1 0.846) on the table alone. It is not a good result -- an 88.0% FPR means the condition flags almost everything regardless of content, with effectively no discriminative value. F1's recall-sensitivity hides this: a reader should weight FPR explicitly rather than rank conditions by F1 alone.

6.7 Directed CoT Is a Confirmed Negative Result, Not an Artifact

At 22.0% detection / 25.0% FPR / F1 0.299 -- computed, per Table 1's caveat, over the 169/200 rows that produced a resolved verdict, not the full 200; the comparison against decoupled's full-200 denominator in Table 2 should be read with this in mind -- and statistically distinct from plain decoupled on every axis ($p<0.05$; Table 2), directed CoT regresses relative to plain decoupled (37.0%/0.0%/0.540) on both detection and FPR. The original rationale -- linking each reasoning step directly into the unsafe-criteria definition, so the model cannot run reasoning as a disconnected track from its verdict the way cot_ablation does -- does not hold up once measured correctly (Section 5.3).

This is not cot_ablation's failure mode recurring. cot_ablation fails by over-triggering (88.0% FPR, flags almost everything). directed_cot fails differently -- moderate FPR (25.0%) paired with low detection (22.0%, worse than plain decoupled's 37.0%) -- so directed reasoning is not simply "more cautious," it is making categorically worse judgment calls in both directions. The likely mechanism: giving the model more surface area to reason over (four steps across five categories) gives it more opportunities to talk itself into an exemption on genuinely harmful prompts, while still occasionally talking itself into a violation on benign ones -- directed reasoning amplifies variance rather than improving calibration. The 15.5% non-completion rate (31/200 unresolved verdicts), even after the parser fix in Section 5.3, is itself worth reporting as a secondary finding: a guard architecture that does not reliably produce a verdict is operationally costly regardless of how it performs on the cases it does resolve. We report this as good evidence against directed CoT as currently structured for a guard task, not a confound or measurement artifact -- it is informative precisely because the rationale was reasonable and still did not pan out.

7. Limitations

- **SWA-amplification untested.** No model in the final set uses genuine sliding-window or sparse attention (Section 4.2.1); the hierarchy-inversion mechanism is tested and appears to work without it, but the amplification hypothesis remains theoretical.
- **Free-tier rate limits.** Groq RPD caps as low as 1,000/day on several models constrained total sample size and required UTC-midnight-aware retry logic (Section 4.5).
- **Conditions 5-8 are not a clean ablation of each other.** The safeguard-family and framing-variant prompts differ qualitatively by design, not on a single isolated variable (Section 4.3).
- **LLM-as-judge reliability, compounded by a discovered failure class.** A reasoning-capable judge model (gpt-oss-120b) silently returned empty output under a small token budget three separate times across this work's evaluation pipeline (judge_quality, judge_refusal, and directed_cot's verdict parser) before being caught (Section 5). A clean-looking results table is not sufficient evidence of correctness when every number traces back to the same small-output-budget judge call; we recommend spot-checking non-empty, well-formed judge output before trusting aggregate metrics in any similar pipeline.
- **Single dataset.** WildGuardMix only; generalization to other harm taxonomies is untested.
- **Sample size and multiple comparisons.** N=100 per class is sufficient to confirm the comparisons in Table 2 at conventional significance levels, but is not large, and Table 2 reports seven simultaneous tests against the same 200-prompt pool without a multiple-comparisons correction (Bonferroni, Holm, or FDR). At an uncorrected $\alpha=0.05$, seven tests carry a non-trivial family-wise false-positive risk. We did not apply a correction here because five of the seven results are extreme ($p<0.001$, with decoupled's FPR literally at 0.0%) and would survive any standard correction; the three borderline results (peer_consensus vs. third_party on detection at $p=0.002$, directed_cot vs. decoupled on detection at $p=0.014$, and peer_consensus vs. third_party on FPR at $p=0.024$) are closer to the threshold and should be read with that caveat in mind rather than treated as equally robust to the $p<0.001$ results.

8. Implications for AMDON

PSNAT-AMDON's guard pipeline is empirically motivated by the corrected comparison in Section 6.3: decoupling's actual value is FPR control, not blanket detection superiority over a well-tuned monolithic model. The framing-axis finding (Section 6.5) additionally motivates AMDON's guard never being asked to self-attribute a persona when classifying, since direct self-judgment framing measurably under-flags relative to distanced framings. The prompting-methodology-transfer finding (Section 6.4) supports using AMDON's pseudo-wrap/hierarchy-inversion guard prompting design even against third-party or future guard models, rather than defaulting to each model's own documented prompting convention.

9. Conclusion

We tested whether decoupling a safety guard from a generation model resolves the structural helpfulness-safety prior conflict, holding the underlying model constant across every role to isolate architecture from capability. The corrected result is more nuanced than a blanket claim of decoupled superiority: a well-tuned monolithic model outperforms plain decoupling on raw detection and F1, and decoupling's defensible advantage is FPR control specifically, recoverable through orthogonal levers -- prompting methodology and self-referential framing -- rather than from architecture alone. Two of those levers are statistically robust under paired bootstrap testing: our guard prompting method transfers to a third-party policy-conditioned model and beats that model's own documented format, and removing self-referential framing produces a real, controllable detection/FPR trade-off. A directed chain-of-thought variant, intended to improve on an earlier undirected-CoT failure mode, is a confirmed negative result rather than an improvement. We also document four evaluation-pipeline bugs uncovered during this work, three sharing a single root cause -- a reasoning-capable judge model silently starving its output under a small token budget -- as a methodological caution for similar LLM-as-judge pipelines. Future work includes a larger-N replication to tighten the confidence intervals reported in Section 6.2, testing against a model with genuine sliding-window attention to evaluate the SWA-amplification hypothesis directly, and integration of the corrected guard architecture into the AMDON pipeline.

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